

Online Judge Problems

- 1 Problem : https://atcoder.jp/contests/abc284/tasks/abc284_b
- 2 Problem : https://atcoder.jp/contests/abc307/tasks/abc307_a
- 3 Problem: https://atcoder.jp/contests/abc368/tasks/abc368_a
- 4 Problem : https://atcoder.jp/contests/abc290/tasks/abc290_a
- 5 Problem : https://atcoder.jp/contests/abc297/tasks/abc297_a
- 6 Problem : https://atcoder.jp/contests/abc308/tasks/abc308_a
- 7 Problem : https://atcoder.jp/contests/abc235/tasks/abc235_b?lang=en
- 8 Problem : https://atcoder.jp/contests/abc286/tasks/abc286_a
- 9 Problem : <https://codeforces.com/contest/1915/problem/C>
- 10 Problem : https://atcoder.jp/contests/abc328/tasks/abc328_b
- 11 Problem : https://atcoder.jp/contests/abc236/tasks/abc236_b?lang=en
- 12 Problem : <https://cses.fi/problemset/task/1083>
- 13 Problem : <https://codeforces.com/contest/1873/problem/B>
- 14 Problem : <https://codeforces.com/contest/1669/problem/B>
- 15 Problem : https://atcoder.jp/contests/abc301/tasks/abc301_b
- 16 Problem : https://atcoder.jp/contests/abc293/tasks/abc293_b
- 17 Problem : https://atcoder.jp/contests/abc330/tasks/abc330_b

18 Problem : https://atcoder.jp/contests/abc280/tasks/abc280_b

20 Problem : <https://codeforces.com/problemset/problem/1878/A>

21 Problem : <https://www.hackerrank.com/contests/101hack39/challenges/equality-in-a-array/problem>

Class Practice

- <https://cses.fi/problemset/task/1083>
- <https://codeforces.com/contest/1915/problem/C>
- https://atcoder.jp/contests/abc308/tasks/abc308_a

- https://atcoder.jp/contests/abc284/tasks/abc284_b
- https://atcoder.jp/contests/abc235/tasks/abc235_b
- https://atcoder.jp/contests/abc307/tasks/abc307_a
- <https://cses.fi/problemset/task/1083>
- https://atcoder.jp/contests/abc236/tasks/abc236_b
- https://atcoder.jp/contests/abc301/tasks/abc301_b
- https://atcoder.jp/contests/abc286/tasks/abc286_a

Array Related Basic Problems

1. Input and Print Array

Take **5 integers** from the user, store them in an array, and print all the elements.

Example:

Input: 10 20 30 40 50

Output: 10 20 30 40 50

2. Print Array in Reverse

Take **N integers** into an array and print them in reverse order.

Example:

Input: 1 2 3 4 5

Output: 5 4 3 2 1

3. Sum of Array Elements

Take **N integers** into an array and calculate the sum of all elements.

Example:

Input: 5 10 15 20

Output: Sum = 50

4. Find the Largest Element

Take **N integers** into an array and find the largest element.

Example:

Input: 12 45 7 89 23

Output: Largest = 89

5. Find the Smallest Element

Take **N integers** into an array and find the smallest element.

Example:

Input: 12 45 7 89 23

Output: Smallest = 7

6. Count Even and Odd Numbers

Take **N integers** into an array. Count how many elements are even and how many are odd.

Example:

Input: 2 7 4 9 6

Output:

Even = 3

Odd = 2

7. Calculate Average

Take **N numbers** into an array and calculate their average.

Example:

Input: 10 20 30 40

Output: Average = 25

8. Search an Element

Take **N integers** into an array. Then take another number and search whether it exists in the array.

Example:

Array: 10 20 30 40 30

Search: 30

Output: Element found

If it does not exist, print Element not found.

9. Count Positive, Negative, and Zero

Take **N integers** into an array and count the number of positive, negative, and zero elements.

Example:

Input: 5 -2 0 8 -7 0

Output:

Positive = 2

Negative = 2

Zero = 2

10. Print Only Elements Greater Than 10

Take **N integers** into an array and print only the elements greater than 10.

Example:

Input: 5 12 8 25 3 15

Output: 12 25 15

11. Copy One Array to Another

Take **N integers** into an array. Copy all elements into a second array and print the second array.

Example:

Input: 10 20 30 40

Output: 10 20 30 40

12. Compare Two Arrays

Take two arrays of the same size and check whether they are equal.

Example 1:

Array 1: 1 2 3 4

Array 2: 1 2 3 4

Output: Arrays are equal

Example 2:

Array 1: 1 2 3 4

Array 2: 1 2 5 4

Output: Arrays are not equal

Day 2

13. Count a Specific Number

Take N integers into an array. Then take a number X and count how many times X appears.

Example:

Array: 2 5 2 8 2 9

X = 2

Output: 2 appears 3 times

14. Find the Second Largest Element

Take N integers into an array and find the second-largest element.

Example:

Input: 10 25 8 40 15

Output: Second largest = 25

15. Find the Second Smallest Element

Take N integers into an array and find the second-smallest element.

Example:

Input: 10 25 8 40 15

Output: Second smallest = 10

16. Separate Even and Odd Elements

Take N integers into an array. Print all even elements first, followed by all odd elements.

Example:

Input: 1 2 3 4 5 6

Output: 2 4 6 1 3 5

17. Replace Negative Numbers with Zero

Take N integers into an array. Replace every negative number with 0, then print the array.

Example:

Input: 5 -2 8 -7 3

Output: 5 0 8 0 3

18. Multiply Every Element by 2

Take N integers into an array. Multiply every element by 2 and print the updated array.

Example:

Input: 1 3 5 7

Output: 2 6 10 14

19. Find the Sum of Positive Elements

Take N integers into an array and calculate the sum of only the positive elements.

Example:

Input: 5 -2 8 -7 3

Output: Sum = 16

20. Find the Index of an Element

Take N integers into an array. Search for a number X and print its index. If it appears multiple times, print the index of its **first occurrence**.

Example:

Array: 10 20 30 40 50 30 40 30

X = 30

Output: Index = 2

Example 2:

Array: 10 20 30 40 50 30 40 30

X = 100

Output: Index = -1

21. Print Elements at Even Indices

Take N integers into an array and print the elements whose indices are even (0, 2, 4, ...).

Example:

Array: 10 20 30 40 50 60

Output: 10 30 50

22. Print Elements at Odd Indices

Take N integers into an array and print the elements whose indices are odd (1, 3, 5, ...).

Example:

Array: 10 20 30 40 50 60

Output: 20 40 60

23. Left Rotate an Array by One Position

Move every element one position to the left. Move the first element to the last position.

Example:

Input: 10 20 30 40 50

Output: 20 30 40 50 10

24. Right Rotate an Array by One Position

Move every element one position to the right. Move the last element to the first position.

Example:

Input: 10 20 30 40 50

Output: 50 10 20 30 40